

Probability Problem set - 1.

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1) $P(\text{Even}) = 2/9, P(\text{odd}) = 1/9$

$\Omega = \{1, 2, 3, 4, 5, 6\}, F = 2^\Omega, P: F \rightarrow [0, 1]$.

generally taken as power set but can also be a subset.

$P(< 4) = P(1) + P(2) + P(3) = 4/9$.

2) $P\left(\bigcup_{i=1}^n A_i^c\right) \leq \sum_{i=1}^n P(A_i^c)$.

$\Rightarrow P\left(\bigcup_{i=1}^n A_i^c\right) \leq \sum_{i=1}^n P(A_i^c)$

Demorgan's law :-

$\Rightarrow P\left(\left(\bigcap_{i=1}^n A_i\right)^c\right) \leq \sum_{i=1}^n [1 - P(A_i)]$

$\Rightarrow 1 - P\left(\bigcap_{i=1}^n A_i\right) \leq n - \sum_{i=1}^n P(A_i)$

$\Rightarrow P\left(\bigcap_{i=1}^n A_i\right) \geq \sum_{i=1}^n P(A_i) - (n-1)$.

3) $P(A) = 0.9, P(B) = 0.9$

(a) i) $A \subseteq B$ (or) $B \subseteq A \Rightarrow B = A$

then $P(A) = 0.9$.

P can't exceed 1 $\rightarrow$ occurs when $A \cup B = \Omega$

$\therefore P(A) \in [0.9, 1]$

(b) $B^c | A$

$$P(B^c | A) = \frac{P(B^c \cap A)}{P(A)}$$

$$= \frac{[0.1, 0.1]}{0.9}$$

$$\Rightarrow [0.1\bar{9}] - [0.10\bar{1}]$$

- false

if B^c and A are disjoint

then $A \cap B^c = \emptyset$

then $P(A \cap B^c) = 0$

$$\text{if } B^c \subset A \text{ then } P(A \cap B^c) = P(B^c) = 0.1$$

(c) $A \cap B$

$$P(A \cap B) = P(A) + P(B) - P(A \cup B)$$

$$\Rightarrow 1.8 - [0.9, 1]$$

$$\Rightarrow [0.8, 0.9]$$

False.

(d) $P(A \cap B) \in [0.8, 0.9]$.

True.

$$(e) P(B^c | A^c) = \frac{P(A^c \cap B^c)}{P(A^c)} = \frac{1 - P(A \cup B)}{0.1} = \frac{1 - [0.9, 1]}{0.1}$$
$$\Rightarrow \left[\frac{0, 0.1}{0.1} \right]$$

- false.

$$= [0, 1]_+$$

(f) from c - True

$$4) a) \Omega = \{BB, BG, GB, GG\}$$

$$P(\{GG\} / \{GB, GG\}) = \frac{P(\{GG\} \cap \{GB, GG\})}{P(\{GB, GG\})} = \frac{P(\{GG\})}{P(\{GB, GG\})}$$

$$= \frac{1/4}{1/2} = 1/2$$

$$b) \Omega = \{BB, BG, GB, GG\}$$

$$P(\{BB\} / \{BG, BB, GB\}) = \frac{P(\{BB\})}{P(\{BG, BB, GB\})} = \frac{1/4}{3/4} = 1/3$$

5)

$$P(A) = P(A \cap B) + P(A \cap C) + P(A \cap B^c \cap C^c) - P(A \cap B \cap C)$$

$$A = A \cap \Omega = A \cap (B \cup B^c) = (A \cap B) \cup (A \cap B^c)$$

$$P(A) = P[(A \cap B) \cup (A \cap B^c)]$$

$$= P(A \cap B) + P(A \cap B^c)$$

$$P(A \cap B^c) = P(A \cap B^c \cap C) + P(A \cap B^c \cap C^c)$$

$$= P(A \cap B) + P(A \cap B^c \cap C) + P(A \cap B^c \cap C^c)$$

$$\Rightarrow P(A \cap B^c \cap C^c) - P(A \cap B \cap C) + P(A \cap B) + [P(A \cap B^c \cap C) + P(A \cap B \cap C)]$$

$$\Rightarrow P(A \cap B) + P(A \cap C) + P(A \cap B^c \cap C^c) - P(A \cap B \cap C)$$

$$6) P(k|m) = p^C (1-p)^{k+m}$$

$$a) \sum_{k=0}^{\infty} \sum_{m=0}^{\infty} p^C (1-p)^{k+m} = 1 \Rightarrow p^C \sum_{k=0}^{\infty} (1-p)^k \sum_{m=0}^{\infty} (1-p)^m = 1$$

$$p^C \left[\frac{1}{1-(1-p)} \right] \left[\frac{1}{1-(1-p)} \right] = 1$$

$$\Rightarrow p^{C-2} = 1 \Rightarrow \boxed{C=2}$$

$$b) p = \frac{1}{4}$$

$$k=m \Rightarrow \sum_{k=0}^{\infty} \sum_{m=0}^{\infty} p^2 (1-p)^{k+m}$$

$$\Rightarrow p^2 \sum_{k=0}^{\infty} (1-p)^{2k} \Rightarrow p^2 \frac{1}{1-(1-p)^2} = \frac{1/16}{1 - \frac{9}{16}} = \frac{1}{7}$$

$$c) \sum_{k=0}^{\infty} \sum_{m=0}^k p^2 (1-p)^{k+m}$$

$$\Rightarrow p^2 \sum_{k=0}^{\infty} (1-p)^k \sum_{m=0}^k (1-p)^m \Rightarrow p^2 \sum_{k=0}^{\infty} (1-p)^k \left[\frac{1-(1-p)^{k+1}}{p} \right]$$

$$\Rightarrow p \sum_{k=0}^{\infty} [(1-p)^k - (1-p)^{2k+1}] \Rightarrow p \left[\frac{1}{p} - \frac{1-p}{1-(1-p)^2} \right]$$

$$\Rightarrow p \left[\frac{1}{p} - \frac{1-p}{p(p+2)} \right] \Rightarrow 1 - \frac{1/4 \left(\frac{3}{4} \right)}{\frac{1}{4} \left(-\frac{3}{4} + 2 \right)} = 1 - \frac{3}{7} = \frac{4}{7}$$

$$\Rightarrow 1 - \frac{3}{7} = \frac{4}{7}$$

d) k is odd

$$\text{then } \sum_{m=0}^{\infty} \sum_{k=0}^{\infty} p^2 (1-p)^{k+m} \Rightarrow k = 2i+1$$

$i = 0, 1, \dots$

$$\Rightarrow \sum_{m=0}^{\infty} \sum_{\substack{i=0 \\ k \text{ is odd}}}^{\infty} p^2 (1-p)^{2i+1+m}$$

$$\Rightarrow \left[\sum_{m=0}^{\infty} (1-p)^m \right] \left[\sum_{i=0}^{\infty} (1-p)^{2i} \right] p^2 (1-p) \Rightarrow \frac{p^2 (1-p)}{p(2-p)}$$

$$\Rightarrow \frac{1-p}{2-p} \Rightarrow \frac{3}{7}$$

e) A: k is even, B: m is even.

$$\Rightarrow P(A) = \sum_{m=0}^{\infty} \sum_{\substack{k=0 \\ k \text{ is even}}}^{\infty} p^2 (1-p)^{k+m}$$

$$k = 2i$$

$$\text{then } \sum_{m=0}^{\infty} \sum_{i=0}^{\infty} p^2 (1-p)^{m+2i}$$

$$\Rightarrow p^2 \left[\sum_{m=0}^{\infty} (1-p)^m \right] \left[\sum_{i=0}^{\infty} (1-p)^{2i} \right]$$

$$\Rightarrow p^2 \left[\frac{1}{p} \right] \left[\frac{1}{p(2-p)} \right]$$

$$\Rightarrow \frac{1}{2-p} = \frac{4}{7}$$

$\rightarrow P(B)$ similar to $P(A)$ and $P(B) = \frac{4}{7}$

$P(A \cap B)$

$$\text{then } P(A \cap B) = p^2 \left[\frac{1}{p(2-p)} \right] \left[\frac{1}{p(2-p)} \right] = \frac{1}{(2-p)^2} = \left(\frac{4}{7} \right)^2$$

$$P(A \cap B) = P(A) \cdot P(B)$$

$\therefore A$ and B are independent.

$$7) \Omega = \{HH, TT, HT, TH\} \quad P(\{H\}) = p$$

$$P(\{HH\} / \{HH, HT\}) \geq P(\{HH\} / \{HH, HT, TH\})$$

$$\Rightarrow \frac{P(\{HH\})}{P(\{HH, HT\})} \geq \frac{P(\{HH\})}{P(\{HH, HT, TH\})}$$

$$\Rightarrow P(\{HH, HT, TH\}) \geq P(\{HH, HT\})$$

$$\Rightarrow 1 - (1-p)^2 \geq p \Rightarrow 2p - p^2 \geq p \Rightarrow -p \leq 0$$

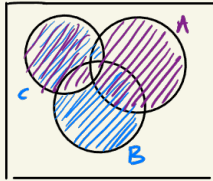
$$\Rightarrow p(p-1) \leq 0 \Rightarrow \underline{p \in [0, 1]}$$

$$8) A \Delta B = (A \cup B) \setminus (A \cap B)$$

$P(A \Delta B) \leq P(A \Delta C) + P(C \Delta B)$ - show that.

$A \Delta C$

$C \Delta B$



if $x \in A \Delta B$ then

$x \in A \Delta C$ 'or' $x \in C \Delta B$ [Infer from diagram]

$$\therefore A \Delta B \subseteq (C \Delta B) \cup (A \Delta C)$$

$$\Rightarrow P(A \Delta B) \leq P((C \Delta B) \cup (A \Delta C)) \rightarrow P(A \Delta B) \leq P(C \Delta B) + P(B \Delta A)$$

$$\boxed{P(A \cup B) \leq P(A) + P(B)}$$

9)

+ve (B)	-ve (B ^c)
0.9	
	0.95

has disease (A)

doesn't have disease (A^c)

$$\Rightarrow P(A) = 0.01$$

$$P(A|B) = ?$$

$$P(B|A) = 0.9, \quad P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)} \Rightarrow \frac{P(B|A) \cdot P(A)}{P(B|A)P(A) + P(B|A^c)P(A^c)}$$

$$P(B|A^c) = 0.95$$

$$P(A|B) = \frac{(0.9)(0.01)}{(0.9)(0.01) + (1-0.95)(1-0.01)} = 0.15384$$

10)

Similar to Monte Hall Problem. See below video for reference

<https://www.youtube.com/watch?v=i7OY7CZUdIY>

11) $P(A_j^c) = 1 - P(A_j)$

B_i : ^{event} paper is found in drawer i .

A_j : event that paper is in drawer j .

$$P(B_i | A_j) = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$$

$$P(A_j | B_i^c) = \frac{P(B_i^c | A_j) \cdot P(A_j)}{P(B_i^c)} = \frac{[1 - P(B_i | A_j)] \cdot P(A_j)}{P(B_i^c)}$$

$$P(A_j^0 | B_i^c) = \frac{p_j^0}{P(B_i^c)}$$

$$\begin{aligned} P(B_i^c) &= \sum_j P(B_i^c \cap A_j^0) \\ &= P(B_i^c \cap A_i^0) + \sum_{j \neq i} P(B_i^c \cap A_j^0) \\ &= P(B_i^c | A_i^0) P(A_i^0) + \sum_{j \neq i} P(B_i^c | A_j^0) \cdot P(A_j^0) \\ &= (1 - d_i^0) p_i^0 + \sum_{j \neq i} p_j^0 \\ &= (1 - d_i^0) p_i^0 + (1 - p_i^0) \end{aligned}$$

12)

$$P(W_1) = \frac{m}{m+n}$$

$$\begin{aligned} P(W_2) &= \frac{m+1}{m+n+1} \left(\frac{m}{m+n} \right) + \frac{m}{m+n+1} \left(\frac{n}{m+n} \right) \\ &= \frac{m^2 + m + mn}{(m+n+1)(m+n)} = \frac{m(m+n+1)}{(m+n+1)(m+n)} \\ &= \frac{m}{m+n} \end{aligned}$$

$$\begin{aligned} P(W_3) &= \frac{m+1}{m+n+1} \left(\frac{m}{m+n} \right) + \frac{m}{m+n+1} \left(\frac{n}{m+n} \right) \\ &= \frac{m}{m+n} \end{aligned}$$

$$\text{III}^y \text{ for } P(\omega_k) : \frac{m}{m+n}$$

$$13) a) P(A|B) > P(A) \Leftrightarrow P(B|A) > P(B)$$

$$P(A|B) \cdot P(B) = P(B|A) \cdot P(A)$$

RHS:

$$P(A|B) > P(A) \Rightarrow \frac{P(B|A) \cdot P(A)}{P(B)} > P(A)$$

$$\Rightarrow P(B|A) > P(B)$$

III^y the reverse implication too.

$$b) P(B^c) > 0 \quad P(A|B) > P(A) \Leftrightarrow P(A|B^c) < P(A)$$

$$P(A|B) > P(A) \Rightarrow P(B|A) > P(B)$$

$$\Rightarrow 1 - P(B|A) < 1 - P(B) \Rightarrow P(B^c|A) < P(B^c)$$

$$\Rightarrow \frac{P(A|B^c) \cdot P(B^c)}{P(A)} < P(B^c) \Rightarrow P(A|B^c) < P(A)$$

III^y prove the reverse implication.

C) A: Treasure at location 1

$$P(A) = \beta \text{ and } P(A^c) = 1 - \beta$$

B: Treasure is found at location 1.

$$P(B/A) = p$$

$$P(A^c/B^c) > P(A^c)$$

$$P(A^c/B^c) = \frac{P(B^c/A^c) P(A^c)}{P(B^c/A^c) P(A^c) + P(B^c/A) P(A)}$$

$$= \frac{1(1-\beta)}{1(1-\beta) + (1-p)\beta} = \frac{1-\beta}{1-p\beta}$$

$$B > pB \Rightarrow 1-\beta < 1-p\beta$$

$$\text{Since } 1-p\beta < 1 \Rightarrow \frac{1}{1-p\beta} > 1 \Rightarrow \frac{1-\beta}{1-p\beta} > 1-\beta$$

$$\Rightarrow P(A^c/B^c) > P(A^c)$$

$$14) q_n = (1 - q_{n-1})p + (q_{n-1})(1-p) \Rightarrow q_n = q_{n-1}(1-2p) + p$$

$$q_n = [q_{n-2}(1-2p) + p](1-2p) + p \Rightarrow q_n = q_{n-2}(1-2p)^2 + p(1-2p) + p$$

$$q_n = q_{n-3}(1-2p)^3 + p(1-2p)^2 + p(1-2p) + p$$

⋮

$$q_n = q_1(1-2p)^{n-1} + p[(1-2p)^{n-1} + (1-2p)^{n-2} + \dots + 1]$$

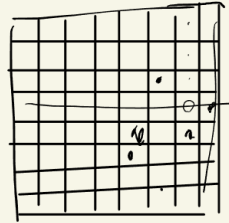
$$q_n = q_1(1-2p)^{n-1} + \left[\frac{1 - (1-2p)^{n-1}}{1 - 1 + 2p} \right] p \Rightarrow q_n = q_1(1-2p)^{n-1} + \frac{1}{2}(1 - (1-2p)^{n-1})$$

$q_1 = 1-p$ [in the first toss we have zero heads therefore we need to get tails to have even heads]

$$\begin{aligned} q_n &= (1-p)(1-2p)^{n-1} + \frac{1}{2}[1-(1-2p)^{n-1}] \\ &= \frac{1}{2}[(2-2p)(1-2p)^{n-1} + 1 - (1-2p)^{n-1}] \\ &= \frac{1}{2}[1 + (1-2p)^n] \end{aligned}$$

15) no. of ways of placing 8 rooks = ${}^{64}C_8$ [all rooks are the same]
safe ways = $8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 8!$

$$\therefore \text{prob} = \frac{8!}{{}^{64}C_8}$$



16) a) probability of birthday of 10 people on different days

$$= \frac{{}^{365}C_{10} \times 10!}{(365)^{10}}$$

b) $\frac{{}^{365}C_n \times n!}{(365)^n} < \epsilon \Rightarrow \frac{365 \times 364 \times \dots \times (365 - (n-1))}{(365)^n} < \epsilon$

$$\Rightarrow \left(1 - \frac{1}{365}\right) \left(1 - \frac{2}{365}\right) \dots \left(1 - \frac{(n-1)}{365}\right) < \epsilon$$

$$1 - x \leq e^{-x} \text{ (hint)}$$

$$\Rightarrow \left(e^{-1/365}\right) \left(e^{-2/365}\right) \dots \left(e^{-(n-1)/365}\right) < \epsilon \text{ [here we are finding a sufficient condition, not the necessary one]}$$

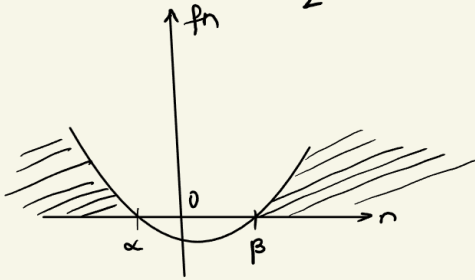
$$\Rightarrow e^{-[(1+2+3+\dots+(n-1))/365]} < \varepsilon \Rightarrow e^{-n(n-1)/730} \leq \varepsilon$$

$$\Rightarrow \frac{n(n-1)}{730} < \ln(\varepsilon) \Rightarrow \frac{n(n-1)}{730} > -\ln(\varepsilon)$$

$$\Rightarrow n(n-1) > -730 \ln(\varepsilon) \Rightarrow n^2 - n + 730 \ln(\varepsilon) > 0$$

$$\Delta = 1 - 4(1)(730 \ln(\varepsilon)) \Rightarrow 1 - 2920 \ln(\varepsilon) > 0 \quad \left[\begin{array}{l} \text{because } \varepsilon \text{ is} \\ \text{a probability} \\ \text{and } \varepsilon \in (0,1) \end{array} \right]$$

$$\text{roots} = \frac{1 \pm \sqrt{1 - 2920 \ln(\varepsilon)}}{2}$$



$$\Delta > 1 \Rightarrow \sqrt{\Delta} > 1$$

$$\text{then } 1 - \sqrt{\Delta} < 0 \text{ and } 1 + \sqrt{\Delta} > 0$$

$$\alpha = \frac{1 - \sqrt{\Delta}}{2}, \quad \beta = \frac{1 + \sqrt{\Delta}}{2}$$

From the figure we can infer that any value of $n < \alpha$ and $n > \beta$ the inequality is satisfied but n can't go

below 0

$$\text{So, } n \geq \left\lceil \frac{1 + \sqrt{1 - 2920 \ln(\varepsilon)}}{2} \right\rceil \quad \text{ceil because } n \text{ is an integer}$$

$\therefore$ For $n \geq \beta$, it is probable that at least 2 birth days fall on same day with " $1 - \varepsilon$ " surity.

example: I want to say with 90% surity $(1 - \varepsilon)$ $1 - \varepsilon = 0.9$.

$$\text{then } n \geq \frac{1 + \sqrt{1 - 2920 \ln(0.9)}}{2} = \frac{18.566}{2} \approx 9.27$$

$$n \geq \lceil 9.27 \rceil \quad n \geq 10$$

$$17]. \quad p_k = p_{k+1}(\beta) + p_{k-1}(1-\beta) \Rightarrow (1-p)p_k + \beta p_k = p_{k+1}(\beta) + p_{k-1}(1-\beta)$$

$$\Rightarrow (p_{k+1} - p_k)\beta = (p_k - p_{k-1})(1-\beta) \Rightarrow \frac{p_{k+1} - p_k}{p_k - p_{k-1}} = \frac{1-\beta}{\beta}$$

$$\Rightarrow \tau_k = p_k - p_{k-1}, \tau_1 = p_1$$

$$\tau_k = \left(\frac{1-p}{p}\right)^{k-1} p_1,$$

$$p_k - p_{k-1} = \left(\frac{1-p}{p}\right)^{k-1} p_1 \Rightarrow p_k = p_1 \left(\frac{1-p}{p}\right)^{k-1} + \left(\frac{1-p}{p}\right)^{k-2} p_{k-2}$$

$$p_k = p_1 \left(\frac{1-p}{p}\right)^{k-1} + p_1 \left(\frac{1-p}{p}\right)^{k-2} + p_1 \left(\frac{1-p}{p}\right)^{k-3} + \dots + p_1$$

$$p_k = p_1 \left[\frac{1 - \left(\frac{1-p}{p}\right)^k}{1 - \left(\frac{1-p}{p}\right)} \right] \rightarrow \textcircled{1}$$

$$p_n = p_1 \left[\frac{1 - \left(\frac{1-p}{p}\right)^n}{1 - \left(\frac{1-p}{p}\right)} \right] \rightarrow \textcircled{2}$$

$$\frac{\textcircled{1}}{\textcircled{2}} = \frac{p_k}{p_n} = \frac{1 - \left(\frac{1-p}{p}\right)^k}{1 - \left(\frac{1-p}{p}\right)^n}, \quad \text{with } p_n = 1$$

$$\Rightarrow p_k = \frac{1 - \left(\frac{1-p}{p}\right)^k}{1 - \left(\frac{1-p}{p}\right)^n}$$

18) a) $A \cap (B \cup C) \quad (\cap B) \cup C$

Case one:

$$x \in A, x \in B, x \notin C$$

$$\text{then } x \in A \cap (B \cup C)$$

$$x \in (\cap B) \cup C$$

Case 2:

$$x \in A, x \notin B, x \in C$$

$$x \in A \cap (B \cup C)$$

$$x \in (\cap B) \cup C$$

Case 3:

$$x \notin A, x \in B, x \in C$$

$$x \notin A \cap (B \cup C)$$

$$x \in (\cap B) \cup C$$

$$\therefore A \cap (B \cup C) \subset (\cap B) \cup C$$

b) $\left(\bigcup_{i=1}^n A_i \right)^c = \left(\bigcap_{i=1}^n A_i^c \right)$

LHS if $x \notin \bigcup_{i=1}^n A_i$, then $x \notin A_i \forall i$

then $x \in \left(\bigcup_{i=1}^n A_i \right)^c$, $x \in A_i^c \forall i$

$$\therefore x \in A_i^c \forall i \Rightarrow x \in \bigcap_{i=1}^n A_i^c$$

$$\therefore \left(\bigcup_{i=1}^n A_i \right)^c \subseteq \bigcap_{i=1}^n A_i^c \rightarrow \textcircled{1}$$

From RHS: if $x \in \bigcap_{i=1}^n A_i^c$ then $x \in A_i^c \forall i$,

then $x \notin A_i \forall i \Rightarrow x \notin \bigcup_{i=1}^n A_i$

$$\Rightarrow x \in \left(\bigcup_{i=1}^n A_i \right)^c$$

this means

$$\bigcap_{i=1}^n A_i^c \subseteq \left(\bigcup_{i=1}^n A_i \right)^c \rightarrow \textcircled{2}$$

From ① and ②

$$\left(\bigcup_{i=1}^n A_i \right)^c = \bigcap_{i=1}^n A_i^c$$

c) A set is countable iff

i) finite

ii) if it is infinite, if there exist a one-to-one mapping from set to $\mathbb{N}$

$$d) A_n = [0, x - \frac{1}{n}]$$

$$A_1 \subset A_2 \subset A_3 \subset A_4 \dots \subset A_n$$

$$\text{then } \bigcup_{i=1}^{\infty} A_i = A_n = [0, x - \frac{1}{n}]$$

$$\lim_{n \rightarrow \infty} \bigcup_{i=1}^n A_i = \lim_{n \rightarrow \infty} A_n$$

$$= [0, \lim_{n \rightarrow \infty} (x - \frac{1}{n})]$$

$$= [0, x]$$

$$\Rightarrow \bigcup_{i=1}^{\infty} A_i = [0, x]$$

- continuity of probability

$$e) B_1 = A_1, B_2 = A_2 / A_1, B_3 = A_3 / A_1 \cup A_2, B_n = A_n / \bigcup_{i=1}^{n-1} A_i$$

$$\bigcup_{i=1}^n A_n = \bigcup_{i=1}^n B_n,$$

$$B_1 = A_1, B_1 \cup B_2 = A_1 \cup (A_2 / A_1) = A_1 \cup A_2$$

$$B_1 \cup B_2 \cup B_3 = A_1 \cup (A_2 / A_1) \cup (A_3 / A_1 \cup A_2)$$

$$= (A_1 \cup A_2) \cup (A_3 / A_1 \cup A_2) = A_1 \cup A_2 \cup A_3$$

By observation,

$$\bigcup_{i=1}^n B_i = \bigcup_{i=1}^n A_i$$

f) A and B are independent.

$$P(A) \cdot P(B) = P(A \cap B)$$

i) A^c and B^c are independent.

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \Rightarrow 1 - P(A^c \cap B^c) = P(A) + P(B) - P(A) \cdot P(B)$$

$$P(A^c \cap B^c) = 1 - (P(A) + P(B)) + P(A) \cdot P(B)$$

$$= 1 - (1 - P(A^c) + 1 - P(B^c)) + (1 - P(A^c))(1 - P(B^c))$$

$$\Rightarrow 1/1 - 1 + P(A^c) - 1 + P(B^c) + 1 - P(A^c) - P(B^c) + P(A^c) \cdot P(B^c)$$

$$= P(A^c) \cdot P(B^c)$$

$$i) P(B/A) = P(B) \Rightarrow 1 - P(B/A) = 1 - P(B)$$

$\Rightarrow P(B^c/A) = P(B^c)$ proved. B^c and A are independent.

9) $\mathcal{F}_1$ is generated by $\mathcal{B}_1 = \{(-\infty, x] \mid x \in \mathbb{R}\}$

i) consider a, b st. $a < b$,

$$(-\infty, a] \text{ and } (-\infty, b] \in \mathcal{F}_1$$

$$\text{then } \underbrace{(-\infty, a]}_A \cap \underbrace{(-\infty, b]}_B \in \mathcal{F}_1 \Rightarrow (a, \infty) \cap (-\infty, b] \in \mathcal{F}_1 \Rightarrow (a, b] \in \mathcal{F}_1$$

ii) we know that $(a, b] \in \mathcal{F}_1$

$$\text{then } (a, b - \frac{1}{n}] \in \mathcal{F}_1$$

$$\text{consider } (a, b - \frac{1}{n}] = A_n \quad \text{then } \bigcup_{i=1}^n A_n = A_n \in \mathcal{F}_1$$

$$\text{then } A_1 \subset A_2 \subset A_3 \dots \subset A_n$$

$$\lim_{n \rightarrow \infty} \bigcup_{i=1}^n A_n \Rightarrow \lim_{n \rightarrow \infty} A_n = (a, b) \in \mathcal{F}_1$$

iii) (a, b)

$$\text{let } A_n = (a - \frac{1}{n}, b] \quad , \text{ then } A_n^c = (-\infty, a - \frac{1}{n}] \cup (b, \infty)$$

$$A_1^c \subset A_2^c \subset \dots \subset A_n^c \in \mathcal{F}_1$$

$$\text{then } \bigcup_{i=1}^n A_n^c = A_n^c \in \mathcal{F}_1$$

$$\lim_{n \rightarrow \infty} \bigcup_{i=1}^n A_n^c \in \mathcal{F}_1 \Rightarrow \lim_{n \rightarrow \infty} A_n^c \in \mathcal{F}_1 \Rightarrow \lim_{n \rightarrow \infty} (-\infty, a - \frac{1}{n}] \cup (b, \infty) \in \mathcal{F}_1$$

$$(-\infty, a) \cup (b, \infty) \in \mathcal{F}_1$$

$$\text{then } [(-\infty, a) \cup (b, \infty)]^c \in \mathcal{F}_1$$

$$[a, b] \in \mathcal{F}_1$$

Since we know $(a, b] \in \mathcal{F}_1$ then $(a, b - \frac{1}{n}] \in \mathcal{F}_1$

$$\text{then } (a, b) \in \mathcal{F}_1.$$

iv) we know that $(a|b)$ and $(a|b) \in f_1$

i.e. $(x, \infty) \in f_1$ and $(x, \infty) \in f_1$

then $\Rightarrow (-\infty, x) \cup (x, \infty) \in f_1$

$\Rightarrow [(-\infty, x) \cup (x, \infty)]^c \in f_1$

$\Rightarrow \{x\} \in f_1$

iii) $(-\infty, x]$ and $(-\infty, x) \in f_1$

(h) $(a|b) \in B_2$

from g we know that $(a|b) \in B_1$

then $f_2 \subset f_1 \rightarrow \textcircled{1}$

now $a = x, b \rightarrow \infty$

then $(x, \infty) \in B_2$

i.e. f_2 can be generated from $(x, \infty) \Rightarrow (-\infty, x] \in f_2$

this means every generator from B_1 can be found in f_2

$\therefore f_1 \subset f_2 \rightarrow \textcircled{2}$

$\textcircled{1}$ and $\textcircled{2} \Rightarrow f_1 = f_2$

$$19) \quad C_0 = [0, 1], \quad C_n = \frac{1}{3} C_{n-1} \cup \left(\frac{2}{3} + \frac{1}{3} C_{n-1} \right)$$

$$a) \quad C_1 = \left[0, \frac{1}{3} \right] \cup \left[\frac{2}{3}, 1 \right]$$

$$C_1 \subset C_0$$

$$C_2 = \left[0, \frac{1}{9} \right] \cup \left[\frac{2}{9}, \frac{8}{9} \right] \cup \left[\frac{6}{9}, \frac{7}{9} \right] \cup \left[\frac{8}{9}, 1 \right]$$

$$C_2 \subset C_1$$

$$C = \bigcap_{n=0}^{\infty} C_n \Rightarrow \lim_{i \rightarrow \infty} \bigcap_{n=0}^i C_n$$

$C = \lim_{i \rightarrow \infty} C_i$ in C_i we have 2^i intervals of size $\frac{1}{3^i}$ [from observation]

$$P(C_i) = \left(\frac{2}{3}\right)^i$$

$$\text{then } P(C) = P\left(\lim_{i \rightarrow \infty} C_i\right) = \lim_{i \rightarrow \infty} P(C_i) = \lim_{i \rightarrow \infty} \left(\frac{2}{3}\right)^i$$

$$P(C) = 0 //$$

b) From C_1 to C_2 , the size of ^{each} interval is reduced by 3 and number of intervals are doubled. By observation, the same happens from C_n to C_{n+1} . After a large n the C_i is just collection of border points. By observation each one of the border points can be mapped to 2012^i and vice versa.

example.

$\frac{1}{3}$ can be mapped to 0022222 - - -

$\frac{2}{3}$ " " " to 0000000 - - -

take few # like this and then apply contour diagonalisation and u get a different sequence which is not a part of the earlier set. This proves that it is uncountable